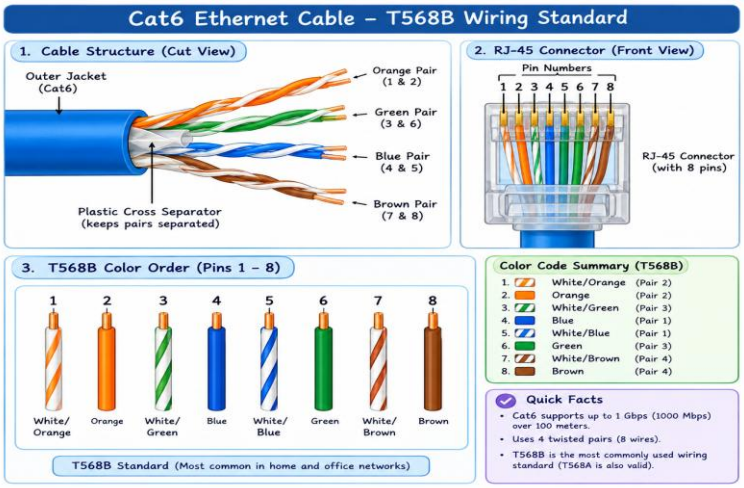
NETWORK CABLE AND CONNECTERS

1.Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.

ANS :



2.List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

ANS:

Three Types of Network Cables and Their Uses

No.	Cable Type	Real-World Device / App	Why It Is Used
1	Cat5e Ethernet Cable	Wi-Fi Router	Connects a router to a computer or other network device and supports reliable wired communication.
2	Cat6 Ethernet Cable	Zomato POS Terminal	Provides a fast, stable wired connection for processing orders and updating payments.
3	Fiber Optic Cable	IPL Streaming Server	Transfers large amounts of data at high speeds over long distances with low signal loss.

3.Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

ANS:

RJ-45 Connector Crimping Process (T568B)

YouTube video: How to Crimp RJ45 Connector – Technical Th1nker. This tutorial demonstrates cable preparation, wire arrangement, crimping, and testing.



YouTube

+1

Step-by-step process

Tools required: Cat6 Ethernet cable, RJ-45 connector, crimping tool, wire stripper, and LAN cable tester.

1. Cut the cable: Cut the Ethernet cable to the required length.
2. Strip the outer jacket: Remove about 2–3 cm of the outer covering carefully without damaging the inner wires.
3. Arrange the wires: Straighten the eight wires and arrange them in T568B colour order: White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.



Aevumix

+1

4. Trim the wires: Cut all eight wires evenly so they fit inside the connector.
5. Insert the wires: Push the wires fully into the RJ-45 connector, keeping the correct colour order.
6. Crimp the connector: Insert the connector into the crimping tool and squeeze firmly to secure the pins and cable.
7. Test the cable: Use a LAN cable tester to check that all eight wires are connected correctly.



CrimpShop

+1

Two common mistakes to avoid

1. Wrong wire order: Always check the T568B colour sequence before inserting the wires.
2. Loose or short wires: Ensure all wires reach the end of the connector and the cable jacket is held securely before crimping.

4.Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

ANS:

Cable and Connector Types for an IPL Gaming Zone

For an IPL screening event, I would choose the following cables and connectors for high-speed and reliable connections.

Cable Type	Connector Type	Use
Cat6 Ethernet Cable	RJ-45	Connect gaming consoles, computers, and routers for a stable, high-speed network.
HDMI Cable	HDMI Connector	Connect gaming consoles or set-top boxes to TVs and projectors for high-quality video and audio.
Fiber Optic Cable	Fiber Optic Connector	Connect network equipment over long distances with high speed and low signal loss.
Power Cable	Power Plug	Supply electricity to TVs, consoles, routers, and other equipment.

Explanation

1. Cat6 Ethernet Cable with RJ-45: It provides a fast and stable wired network connection, reducing lag during online gaming and streaming.
2. HDMI Cable: It transmits high-quality video and audio from consoles or set-top boxes to large screens.
3. Fiber Optic Cable: It is suitable for connecting network equipment over long distances and carrying large amounts of data.
4. Power Cable: It provides a reliable power supply to all devices.

**5.Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.

Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.**

ANS:

Network Cable and Connector in Use

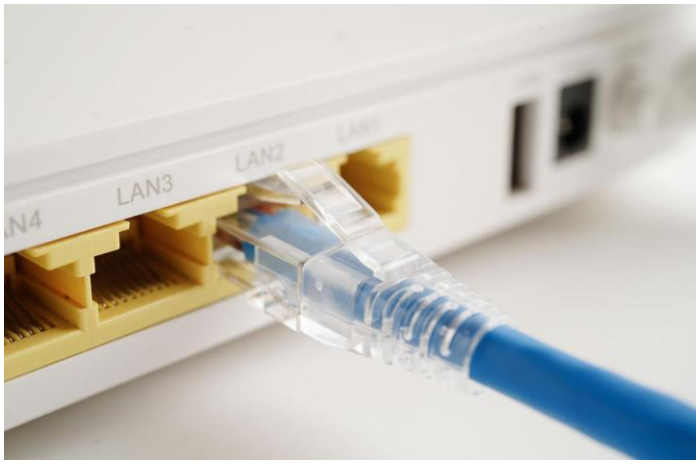


Photo: Ethernet cable connected to a Wi-Fi router.

Description

- Cable type: Cat5e/Cat6 Ethernet cable.
- Connector type: RJ-45 connector.
- Devices connected: Wi-Fi router and a computer, switch, or other network device.
- Purpose: It transfers data between devices and provides a stable wired network connection.
- Advantage: It offers reliable connectivity and generally lower latency than Wi-Fi.

TP-Link

+1

Conclusion: This Ethernet cable is commonly used in homes, colleges, and offices to connect network devices to a router or switch.